

Bhav-Net: Multilingual Dual-Space Graph Transformers for Cross-Lingual Antonym Detection

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Abstract

Antonym detection poses unique challenges as antonymous words often appear in similar contexts despite expressing opposite meanings. We present Bhav-Net, a novel dual-space architecture that addresses this challenge through separate semantic projections for synonymy and antonymy relationships. Our approach combines language-specific BERT encoders with graph transformer networks to model word pairs as bidirectional graphs, enabling higher-order relational reasoning. The dual-encoder design creates distinct semantic spaces where synonymous pairs cluster in synonym space while antonymous pairs exhibit high similarity in antonym space. We evaluate Bhav-Net across seven languages using real-world datasets from WordNet and ConceptNet, demonstrating consistent performance improvements over baseline methods. The architecture achieves strong cross-lingual generalization while respecting language-specific characteristics, with particularly notable improvements on verb and adjective antonym detection tasks.

1 Introduction

Detecting semantic relationships between words remains a fundamental challenge in natural language processing. While synonyms share semantic similarity and distributional patterns, antonyms present a unique paradox: they share semantic domains but express opposite meanings. Words like "hot" and "cold" both relate to temperature yet represent opposing concepts, often appearing in similar linguistic contexts despite their contrasting semantics.

This distributional challenge has led most computational approaches to focus on synonym detection, where contextual similarity aligns with semantic relationship. Antonym detection, however, requires more sophisticated modeling that can distinguish between shared semantic domains and oppositional relationships within those domains.

Recent advances in multilingual language models offer opportunities for cross-lingual semantic understanding, but few approaches explicitly model the dual nature of semantic relationships across diverse languages. Existing multilingual methods typically treat all semantic relationships uniformly, leading to poor separation between synonymous and antonymous pairs.

We introduce Bhav-Net,¹ a multilingual dual-encoder architecture that tackles these challenges through explicit dual-space modeling. Our key insight is that synonyms and antonyms require fundamentally different representational spaces to be properly distinguished.

Contributions: (1) A dual-space projection mechanism that learns distinct patterns for synonym and antonym relationships; (2) Graph transformer architecture for higher-order relational reasoning over word pairs; (3) Comprehensive multilingual evaluation across seven languages demonstrating robust cross-lingual performance; (4) Analysis of architectural design principles that enable effective semantic relationship modeling.

2 Related Work

Early work in semantic relationship detection relied on lexical resources and pattern-based approaches (??). While effective for synonym identification, these methods struggled with antonyms due to their paradoxical nature of semantic similarity with oppositional meaning.

Distributional approaches emerged with the advent of word embeddings (??). These methods excel at capturing semantic similarity but inherently conflate synonyms and antonyms, as both appear in similar contexts. Several approaches have attempted to address this limitation by incorporating

¹Named after the Sanskrit word "bhav" meaning emotion or feeling, reflecting the model's ability to capture semantic relationships.